

GPU-Based Path Tracing

PS Visual Computing

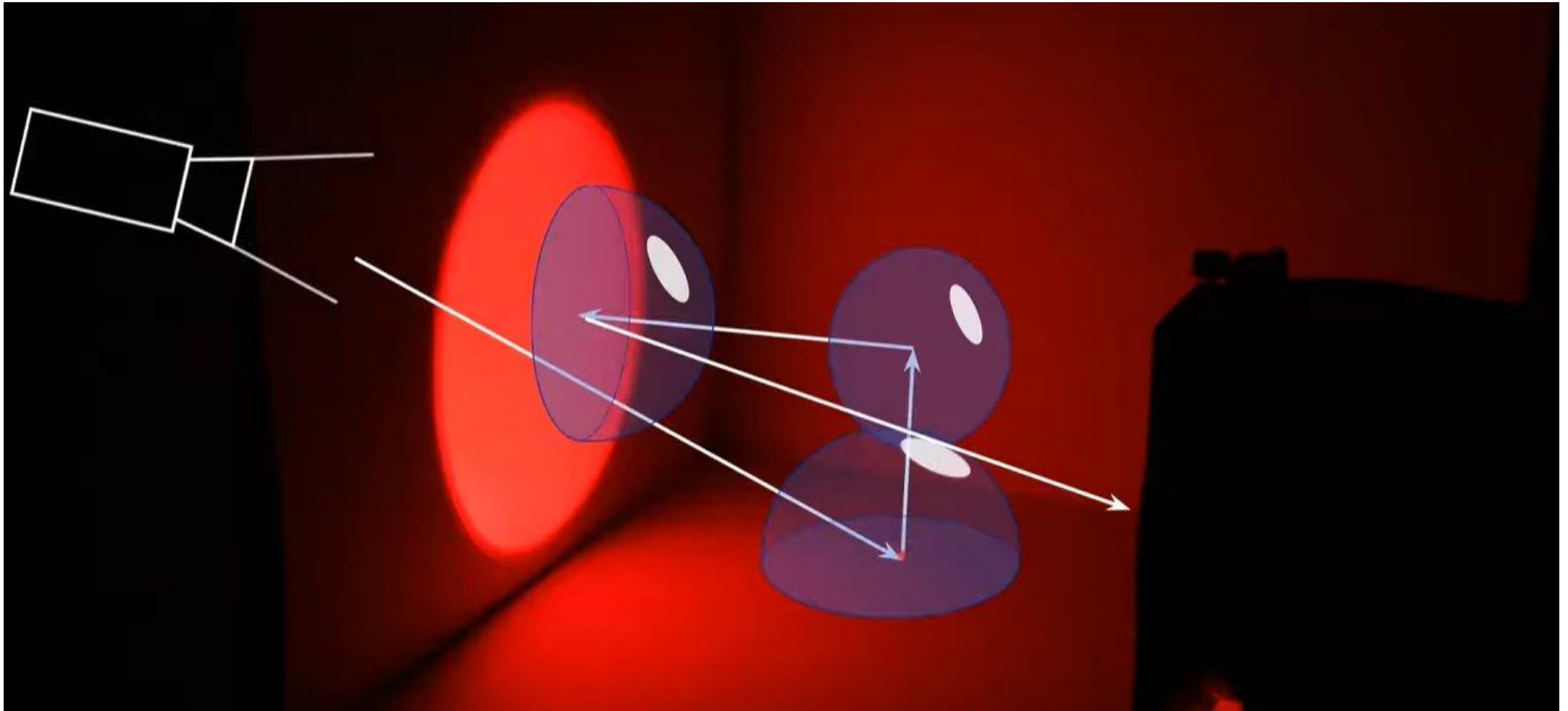
Okan Güclü

What is Path Tracing?

- **Ray Tracing**
 - Shoot rays into the scene, report on hit objects
 - Bounce into new directions, stop after some time
 - No claim to authenticity
- **Path Tracing**
 - Theoretically infinite bounces (high quality)
 - Approximates actual light transport (physically based)
- My Path Tracer is an **unbiased** path tracer



Cyberpunk 2077



Theory behind it

■ The Rendering Equation [1]

$$L_e(x, v) = E(x, v) + \int_{\Omega} f_r(x, \omega \rightarrow v) L_i(x, \omega) \cos(\theta_x) d\omega$$

Diagram illustrating the components of the Rendering Equation:

- Material modelled by the BRDF**: Points to $f_r(x, \omega \rightarrow v)$.
- Light from direction ω** : Points to $L_i(x, \omega)$.
- Solid angle**: Points to $d\omega$.
- Light going in direction v** : Points to $L_e(x, v)$.
- Light emitted from x in direction v** : Points to $E(x, v)$.

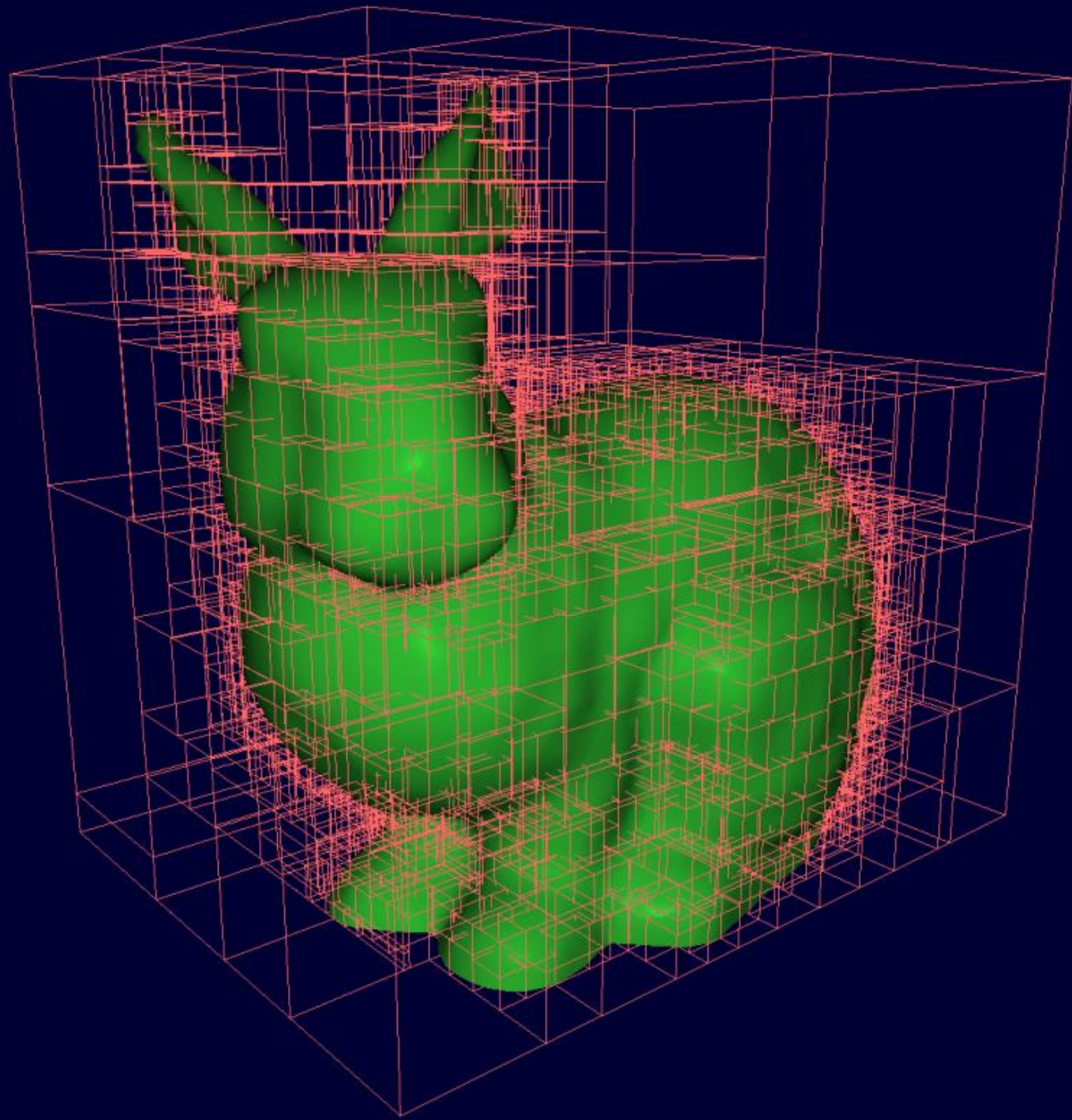
Visual aids:

- A blue sphere representing a material point x with a light source (white circle) and a viewer (white circle).
- Two diagrams illustrating solid angle: a vertical grid of lines and a perspective view of a grid of lines.

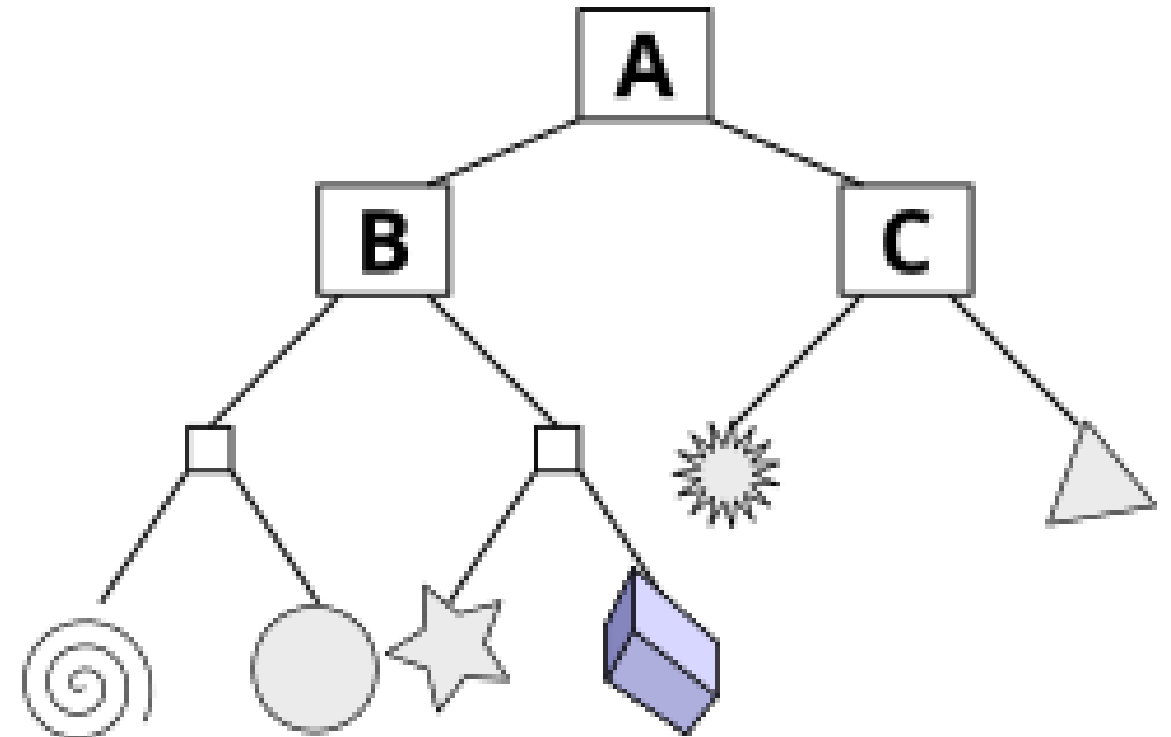
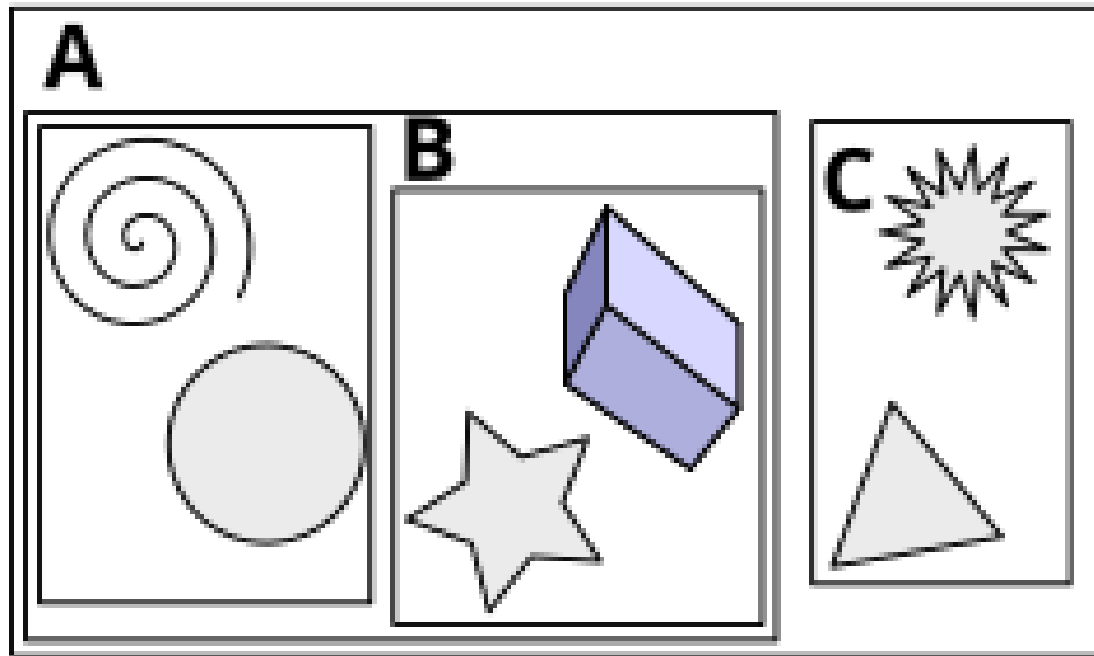
[2]

Infinitely dimensional, not possible to write an analytical solution for a particular scene

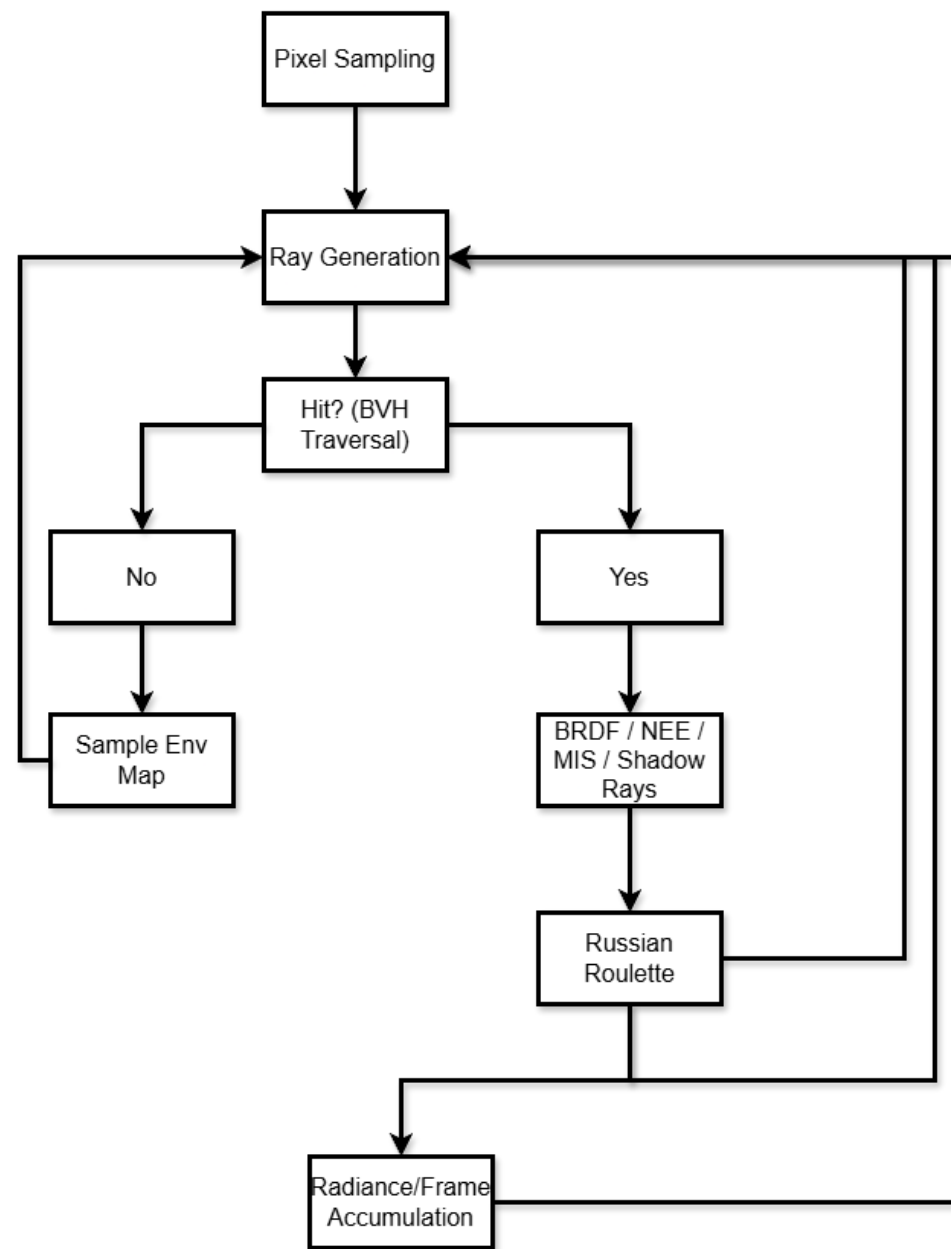
MONTE CARLO



BVH (Bounding Volume Hierarchy)



Path to victory





February 26





February 26



February 26

Github Repo: <https://github.com/AAstroPhysiCS/PathTracer>

